SUCHITHRA BALAKRISHNAN

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EDUCATION

Worcester Polytechnic Institute (WPI) Worcester, MA
Master of Science in Data Science May '18

Anna University Chennai, India
Bachelor of Engineering in Computer Science and Engineering May '16

SKILLS

Languages & Libraries: Python, R, SAS, MATLAB, SQL, Pandas, NumPy
Machine Learning & AI: Scikit-learn, XGBoost, LightGBM, Random Forest, TensorFlow/Keras, SpaCy, NLTK, anomalize, LangChain, LanGraph
Data Engineering & Cloud: AWS (EMR, Athena, S3), Snowflake, MySQL, PostgreSQL, MongoDB, dbt, Airflow
Visualization & Reporting: Matplotlib, Seaborn, Plotly, Power BI, Tableau, Google Analytics, Excel
MLOps & DevOps: Git, Github Copilot, Claude Code, REST API deployment

WORK EXPERIENCE

Senior Manager, Data Science, Fidelity Investments, MA Dec '21-Present

- Developing a **context-aware, rules-based vulnerability prioritization framework** leveraging **dbt, Python, and Snowflake** to integrate and transform data from security scan results, threat intelligence feeds, asset inventories, and the National Vulnerability Database (NVD) to enable systematic and data-driven prioritization of security vulnerabilities across the organization
- Architected a **proof-of-concept insider threat and data loss prevention solution** by querying and aggregating large-scale user activity logs in **Snowflake and Python**, applying supervised and unsupervised machine learning techniques including **XGBoost and Autoencoders** to detect anomalous user behavior and mitigate data loss risks
- Designed and championed **data science model development, deployment standards, and governance best practices**, enabling cross-functional data science teams to adopt sound methodologies, standardized tooling, and reproducible workflows through hands-on examples and targeted training
- Mentored and supported the technical growth of **junior data scientists and interns**, providing guidance on rigorous analytical thinking, modeling strategies, and best practices to elevate the quality and robustness of their work
- Led two semester-long Fidelity Investments-sponsored capstone projects for 3 years in partnership with **Boston University**, guiding approximately 15 students per cohort on cutting-edge topics including **evaluating LLMs for performance efficiency across model sizes** and **assessing knowledge base quality for RAG systems**
- Collaborated with the **Fraud Prevention team** to evaluate and enhance digital transaction ML models, applying rigorous analytical frameworks to improve model robustness and fraud detection capabilities
- Built **analytics solutions leveraging code commit and DevOps data** to identify patterns in development workflows, delivering actionable frameworks and training programs to help developers build more robust, higher quality code and accelerate time to market
- Developed an **XGBoost classifier model to reduce false positives in employee trade compliance alerts by ~20%**, ensuring employees are flagged accurately for trading without proper pre-approval while significantly reducing alert fatigue for compliance teams
- Improved **international Anti-Money Laundering (AML) transaction ML model performance by ~10%** in partnership with the AML team, enhancing the model's ability to detect suspicious financial activity
- Conducted **independent evaluation of a third-party AI/ML vendor solution**, performing comprehensive technical and strategic analysis and presenting findings and recommendations to business stakeholders and leadership to inform enterprise technology investment decisions
- Built an **MVP ML model for IT risk prediction**, improving change management ticket risk classification accuracy by **~30%** to enable more effective prioritization and resolution

Data Scientist/Statistician II, MAPFRE Insurance, MA Oct '19-Dec '21

- Led development of multiple production-grade **predictive models spanning regression, decision trees, and neural networks** leveraging customer demographics, claims, and transactional data to drive measurable improvements in **fraud detection, claims handling, and customer retention**
- Spearheaded the **first-ever REST API-based ML model deployment** using an internal MLOps platform within the **AWS ecosystem** for fraud detection, architecting the end-to-end pipeline from model development to production deployment
- Led the **migration and modernization of a legacy SAS fraud detection model to AWS** (EMR, Athena, S3), collaborating with MLOps infrastructure team to deliver significant gains in **processing speed, deployment efficiency, component reusability, and end-to-end model traceability**
- Conducted independent **evaluation of third-party Data Science vendor solutions**, performing comprehensive technical and strategic analysis and presenting findings and recommendations directly to **senior leadership** to inform enterprise-wide technology investment decisions
- Architected a **statistical data drift monitoring framework in Python**, engineering dynamic threshold-based alerting across key KPIs for production models to enable proactive detection of data quality degradation and model performance deterioration before business impact
- **Mentored and onboarded** junior data scientists and data analysts, providing guidance across technical skills, tooling, and best practices, while fostering a culture of continuous learning and knowledge sharing within the data science organization

Data Analyst II, MAPFRE Insurance, MA Jun '18-Oct '19

- Reduced manual processing time by an estimated **20% by engineering an automated text extraction pipeline** in Python using **pdf2image and Tesseract OCR** to process thousands of scanned incident and police report PDFs, enabling efficient extraction of key data points for downstream predictive modeling and analytics
- Developed a **SpaCy NLP proof-of-concept model** to automatically extract and classify accident severity indicators from claims representative incident reports, successfully presenting the model's business value to stakeholders and receiving strong organizational endorsement for its potential to streamline claims processing workflows
- Architected and maintained scalable **SAS data preparation and integration pipelines** across multiple projects, engineering and cleaning data from hundreds of source system tables to power fraud detection, claims handling, and customer retention predictive models
- Designed and automated **end-to-end model monitoring and performance evaluation frameworks** in Python, SAS, Excel and Power BI across multiple production models, tracking critical metrics and delivering actionable performance insights to fraud, claims, and customer retention business units on a monthly and quarterly basis

Consumer Analytics and Insights Data Science Co-op, Fidelity Investments, RI Jan '18-Jun '18

- Developed an **anomaly detection model in R** using the `anomalize` package to monitor and detect data drift, missing data, and unexpected shifts in input features, like consumer demographics and transactional data, across production ML models, engineering an automated alerting pipeline to enable proactive triaging and corrective action by model owners
- Researched and synthesized global **AI/ML governance best practices** into a comprehensive proposed framework covering data and model governance standards across the full ML lifecycle, presenting findings and recommendations to peer data science teams and their managers across multiple AI/ML business units for evaluation and adoption
- Conducted systematic **ML model audits** across tens of production models by designing and maintaining a model inventory tracking system, collaborating with model owners and business stakeholders, and analyzing performance dashboards to assess model health, relevance, and usage, to identify stale or underperforming models and recommending retirement or retraining to maintain a **reliable and trustworthy production model ecosystem**

Graduate Qualifying Project Data Science Intern, MAPFRE/Verti Insurance, MA Jan '18-May '18

- Developed and implemented customer segmentation models using **K-Means and Hierarchical Clustering in Python**, leveraging customer demographic and web behavioural data from their digital quote platform to identify high-value customer segments and drive conversion from quote to customer

- Engineered end-to-end **data tracking and monitoring pipelines using Google Analytics**, mapping and analyzing customer behavior across the digital insurance quoting to identify drop-off points, enabling data-driven targeting strategies aimed at increasing quote volume to high-conversion audience segments

Data Science Intern, Pacific Northwest National Laboratory (U.S. DOE)**,** WA Jun '17-Aug '17

- Developed and validated **Python-based regression models** for ZIP load forecasting in power distribution systems, achieving over 80% predictive accuracy across Washington State's power grid, as part of the GridAPPS-D Volt-Var Optimization (VVO) application
- Engineered a **modular, reproducible MATLAB backend** for the Grid Project Impact Quantification (GridPIQ) tool, designing end-to-end data workflows to compute transmission and distribution line losses, serving as a critical input for statewide grid impact assessments
- Architected and deployed **internal and external data scraping and processing pipelines** in **MATLAB and Python**, building predictive models for the Conservation Voltage Reduction (CVR) and Electrification modules within GridPIQ, quantifying key energy impact metrics including electricity demand reduction and emissions savings attributed to voltage optimization and electric vehicle adoption across the state

Intern, Ravaani, LLC**,** Chennai Jun '15

- Designed and implemented an **OLAP schema-based MySQL data warehouse** to store and structure large-scale sales and inventory data, enabling efficient multidimensional analytics and supporting data-driven business decision making
- Developed an **ETL pipeline using Pentaho Data Integration**, engineering automated data extraction and transformation workflows to migrate raw data from a content management system into a structured analytical database, ensuring data consistency, integrity, and scalability across the pipeline
- Developed a **Mondrian OLAP schema** and configured the **Pentaho BI Server** to load, aggregate, and serve data insights through a streamlined reporting layer, enabling business stakeholders to efficiently query and analyze large-scale sales and inventory trends

CERTIFICATIONS

- Product Management Certification by Cornell University (Ongoing)
- Inquiry-Driven Leadership Certification by MIT (2025)
- Leadership Essentials Certification by Cornell University (2024)
- Amazon Web Services Certified Machine Learning – Specialty (2021)
- Amazon Web Services Certified Solutions Architect – Associate (2021)
- EMC Data Science Associate (2015)
- Microsoft Certified Professional in MS SQL Server 2012 Databases (2014)